

Project Summary

Objective

- Analyze financial and trading data to understand market trends, investor sentiment, and historical performance patterns.
- Explore correlations between the Fear & Greed Index and historical market data.

Data

- `fear_greed_index.csv`: Daily Fear & Greed Index values representing market sentiment.
- `historical_data.csv`: Historical trading data including prices, volumes, and key market indicators.

Analysis

1. **Data Cleaning**
 - Handled missing values and ensured consistent formats across datasets.
2. **Exploratory Data Analysis (EDA)**
 - Visualized trends in Fear & Greed Index over time.
 - Compared market indices and sentiment to detect patterns.
3. **Correlation Analysis**
 - Measured correlation between market indicators and investor sentiment.
 - Identified periods of high greed vs high fear and their effect on market performance.
4. **Insights**
 - Market sentiment (Fear & Greed Index) often precedes significant market movements.
 - Certain historical patterns repeat during high fear or greed periods.
5. **Outputs**
 - Plots, charts, and graphs saved in `outputs/` folder for easy reference.

Conclusion

- The analysis shows that **investor sentiment is a strong indicator of market trends**.
- Historical patterns in the Fear & Greed Index can help anticipate potential market behavior.
- Further modeling could include predictive analytics or machine learning for trend forecasting.